

KUBO



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
12 DAYS (x2 LEVELS) (1
HOUR PER SESSION)**

**MODE:
OFFLINE**

KUBO Robotics is a hands-on, screen-free coding program that introduces students to computational thinking, logical reasoning, and problem-solving using the KUBO robot and TagTiles. Through interactive challenges and mission-based activities, learners explore sequencing, functions, subroutines, loops, and Coding+ while developing essential STEM skills.

COURSE CURRICULUM

LEVEL 1: KUBO Robotics Fundamentals

Session 1: Introduction to Routes & Directional Commands

- Introduction to KUBO robot and TagTiles
- Understanding directions and movement commands
- Learning route creation through sequencing
- Executing simple navigation tasks
- Building confidence with basic robot movement

Session 2: Route Planning & Memory

- Understanding route planning strategies
- Creating efficient navigation paths
- Developing sequence memorization skills
- Comparing multiple route solutions
- Solving shortest-path challenges

Session 3: Introduction to Functions

- Understanding reusable command sequences
- Learning the concept of functions
- Converting routes into reusable blocks
- Reducing repetitive commands
- Building efficient navigation programs

Session 4: Functions & Logical Storytelling

- Applying functions in structured tasks
- Understanding sequence interpretation
- Building story-based navigation challenges
- Developing logical reasoning skills
- Creating functional route narratives

Session 5: Subroutines

- Introduction to subroutines and modular thinking
- Understanding structured execution
- Combining multiple functions together
- Simplifying complex navigation tasks
- Building multi-step robotic programs

Session 6: Subroutines & Debugging

- Executing programs using multiple subroutines
- Identifying logical errors in routes
- Understanding debugging techniques
- Refining navigation accuracy
- Improving program efficiency

Session 7: Loops

- Introduction to repetition and loops
- Identifying repeatable movement patterns
- Reducing command complexity
- Understanding loop execution
- Creating optimized navigation routes

Session 8: Loops & Competitive Challenges

- Applying loops in mission-based activities

- Improving execution efficiency
- Solving speed and optimization challenges
- Comparing different coding approaches
- Preparing for competition-style tasks

Session 9: IRO Challenge 1

- Understanding challenge objectives
- Applying route planning techniques
- Using functions and loops effectively
- Navigating within mission constraints
- Completing structured robotics missions

Session 10: IRO Challenge 2

- Implementing obstacle-aware navigation
- Applying logical decision-making
- Optimizing movement sequences
- Solving task-based challenges
- Demonstrating efficient route execution

Session 11: IRO Challenge 3

- Planning multi-step missions
- Combining functions, subroutines, and loops
- Managing complex navigation requirements
- Testing mission accuracy
- Executing structured challenge solutions

Session 12: IRO Challenge 4

- Applying all learned concepts in a final challenge
- Optimizing routes and execution logic
- Completing integrated robotics missions
- Demonstrating problem-solving skills
- Showcasing end-to-end computational thinking

LEVEL 2: Advanced KUBO Robotics

Session 1: Advanced Route Observation

- Understanding orientation and directional behavior
- Analyzing robot movement patterns
- Predicting route outcomes
- Improving navigation accuracy
- Solving advanced observation challenges

Session 2: Advanced Functions

- Revisiting function-based programming

- Creating complex functional sequences
- Testing and debugging functions
- Improving code organization
- Building efficient navigation systems

Session 3: Subroutines Integration

- Combining multiple functions and subroutines
- Understanding modular programming concepts
- Structuring large navigation tasks
- Managing program flow effectively
- Executing integrated solutions

Session 4: Loops Optimization

- Identifying repetitive structures
- Applying advanced loop logic
- Reducing program length
- Improving execution efficiency
- Solving optimization challenges

Session 5: Coding+ Fundamentals

- Introduction to Coding+ TagTiles
- Understanding distance, speed, time, and direction
- Controlling advanced movement parameters
- Exploring precision navigation
- Building controlled motion sequences

Session 6: Advanced Functions with Coding+

- Designing complex functional systems
- Combining Coding+ parameters with functions
- Improving navigation precision
- Testing advanced movement logic
- Building structured robotic solutions

Session 7: Advanced Subroutines with Coding+

- Creating layered function systems
- Integrating Coding+ commands into subroutines
- Managing complex program structures
- Executing multi-stage tasks
- Refining program performance

Session 8: Loops & Recursive Functions

- Understanding advanced repetition concepts
- Applying loops within functional systems
- Exploring recursive logic

- Implementing continuous execution patterns
- Solving advanced coding challenges

Session 9: IRO Challenge 1

- Applying structured logic in competitive missions
- Managing task constraints effectively
- Using advanced programming concepts
- Testing solution accuracy
- Completing challenge objectives

Session 10: IRO Challenge 2

- Executing multi-condition task sequences
- Applying optimization techniques
- Improving navigation efficiency
- Solving advanced robotics missions
- Demonstrating logical decision-making

Session 11: IRO Challenge 3

- Integrating navigation and obstacle management
- Combining delivery and movement logic
- Managing complex mission requirements
- Testing and refining solutions
- Completing advanced challenge scenarios

Session 12: IRO Challenge 4

- Applying all Level 2 concepts in a final mission
- Optimizing coding strategies and execution
- Solving real-world inspired robotics challenges
- Demonstrating advanced computational thinking
- Showcasing complete mission performance

CLASS STRUCTURE (STANDARD FOR ALL SESSIONS)

- Clear explanation of session objectives
- Guided hands-on activities using KUBO and TagTiles
- Real-time testing, troubleshooting, and debugging
- Collaborative problem-solving exercises
- Discussion on computational thinking and real-world applications
- Challenge-based learning and performance evaluation